

Robust LOS Rate Estimation for Strapdown SAL Seekers under Specular Reflections: Experimental Characterization and Huber-Based Filtering

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Abstract

This paper presents a robust guidance filter designed to estimate the line-of-sight (LOS) rate of guided missiles equipped with strapdown Semi-Active Laser (SAL) seekers. While strapdown seekers offer significant structural simplicity and cost advantages, their estimation accuracy is severely degraded by non-Gaussian measurement noise, particularly specular reflections that introduce large, outlier-type bias errors. To overcome the inherent limitations of a conventional Extended Kalman Filter (EKF)-based Guidance Filter (GF) under these conditions, a Huber-based Guidance Filter (HGF) is proposed. By integrating the Huber cost function into the recursive filtering framework, the proposed HGF effectively suppresses the influence of outliers while maintaining nominal estimation accuracy under Gaussian noise conditions. Outdoor experiments using

vehicle surfaces with different reflective characteristics are conducted to characterize the occurrence and qualitative behavior of specular-reflection-induced angular disturbances. Motivated by the observed finite-duration, bias-like behavior, controlled angular-bias profiles with prescribed magnitudes and durations are employed to assess filter robustness; these profiles are not statistically identified from the outdoor measurements. Numerical simulation results demonstrate that the proposed HGF improves LOS-rate estimation accuracy and intercept performance compared with the conventional GF, a H_∞ guidance filter, and an improved maximum correntropy Kalman filter (IMCKF) benchmark under the considered disturbance conditions.

Keywords: Guidance Filter, Line-of-Sight Rate Estimation, Strapdown Semi-Active Laser Seeker, Specular Reflection, Huber Filter

1 Introduction

In the development of modern precision-guided munitions (PGMs), particularly for surface-to-surface and air-to-ground engagements, strapdown Semi-Active Laser (SAL) seekers have been increasingly adopted. Compared to conventional gimbale infrared (IR) platforms, strapdown SAL seekers offer distinct advantages, including mechanical simplicity, reduced weight, and lower manufacturing costs. By eliminating complex mechanical gimbal assemblies, these seekers facilitate the miniaturization of the weapon system and significantly lower development and production costs, making them well-suited for high-volume deployment.

Beyond economic advantages, the strapdown architecture provides significant benefits regarding guidance loop dynamics. The absence of mechanical gimbals removes associated mechanical lags, thereby enabling high-bandwidth signal processing. This characteristic is particularly effective in mitigating the stability issues associated with the “parasite loop,” a phenomenon often observed in high-gain guidance systems that can lead to performance degradation [1–7]. Consequently, strapdown architectures are increasingly preferred for tactical missiles where rapid response times and robust tracking stability are critical.

The operation of a SAL seeker relies on the detection of reflected laser energy from a designated target using a four-quadrant photodiode detector [8–11]. Ideally, the target surface produces *Lambertian reflections*, resulting in a uniform energy distribution that allows measurement noise to be modeled as a Gaussian process. However, in practical engagement scenarios, targets often possess surfaces with high reflectivity, such as glass windows or polished metallic bodies. These surfaces induce specular reflections, which cause a non-uniform concentration of laser energy on the detector plane. Such concentrations result in significant centroid shifts, producing large, outlier-type measurement biases that violate the Gaussian noise assumption inherent in standard estimation algorithms.

To estimate the line-of-sight (LOS) rate required for guidance commands, a variety of filtering architectures utilizing SAL seeker measurements and onboard inertial sensors have been investigated. Early approaches employed alpha-beta filters to estimate

the look-angle rate, which was then fused with inertial rate data [3–5]. Subsequent advancements focused on Kalman Filter (KF) formulations to enhance estimation precision. For instance, Kim et al. [12] utilized LOS unit vectors as state variables within an inertial-frame model, while later works extended this to body-frame models incorporating state-delay compensation [7]. Other studies have structured LOS rate dynamics by including the range-rate-to-range ratio as a state variable [13–18], or addressed sensor asynchronicity and time delays through relative kinematics-based asynchronous filtering [19].

Despite these architectural advancements, a fundamental limitation persists: the majority of these methods are grounded in the ℓ_2 -norm cost function, aiming to minimize the Mean Squared Error (MSE). While optimal under the assumption of Gaussian noise, ℓ_2 -norm-based estimators are inherently fragile in the presence of non-Gaussian disturbances. As the quadratic cost function disproportionately penalizes large residuals, even a single significant outlier—such as those induced by specular reflections on a target surface—can severely skew the estimated mean, leading to filter instability and degraded robustness.

To mitigate the sensitivity of standard estimators to such non-Gaussian outliers, robust M-estimation and filtering techniques, particularly those employing the Huber cost function, have been proposed [20–26]. Other robust filtering approaches, such as maximum-correntropy-based Kalman filtering, have also been studied for non-Gaussian noise environments [27]. Unlike H_∞ filtering, which minimizes the worst-case estimation error [28], the Huber filter offers a hybrid approach: it behaves as an ℓ_2 -norm estimator for small residuals (nominal Gaussian noise) and transitions to an ℓ_1 -norm estimator for large residuals (outliers). This adaptive characteristic allows the Huber-based framework to maintain high accuracy under normal conditions while effectively suppressing the destabilizing effects of impulsive noise, making it an ideal candidate for handling the mixed-reflection characteristics of SAL seekers.

This paper investigates robust LOS-rate estimation for strapdown SAL seekers under specular-reflection conditions. The contributions of this study are threefold. First, we experimentally characterize the occurrence and qualitative behavior of specular-reflection-induced angular disturbances through outdoor experiments using a test vehicle. These experiments examine surface-dependent, bias-like angular deviations associated with different vehicle materials and mixed-reflectivity patterns. Second, a Huber M-estimation framework is adapted to the recursive LOS-rate estimation process used for strapdown SAL seeker guidance. Third, the resulting guidance filter is evaluated through closed-loop engagement simulations using estimation and intercept-performance metrics, with comparisons against the conventional EKF-based GF, the H_∞ Guidance Filter (HinfGF), and an improved maximum correntropy Kalman filter (IMCKF) benchmark under the presence of specular-induced outliers. The remainder of this paper is organized as follows. Section 2 defines the engagement geometry and the system models. Section 3 presents the experimental characterization of specular-reflection-induced angular disturbances based on outdoor experiments. Section 4 describes the design and implementation of the proposed robust HGF. Simulation results and performance comparisons are discussed in Section 5, followed by conclusions in Section 6.

2 System Modeling and Problem Formulation

This section establishes the mathematical framework for the missile-target engagement geometry and the measurement dynamics of the strapdown SAL seeker. We focus on the kinematic coupling between the body-fixed sensor measurements and the inertial guidance frame, which is critical for understanding the impact of seeker noise.

2.1 Engagement Geometry

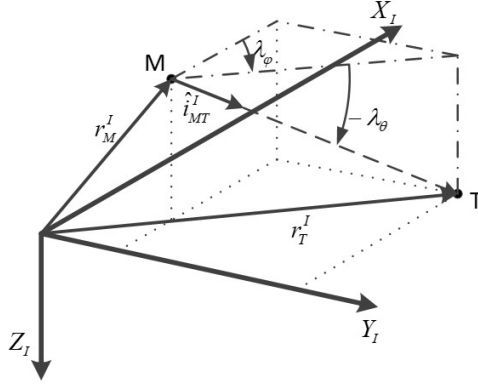


Fig. 1 Three-dimensional engagement kinematics and coordinate system definitions.

We consider a three-dimensional engagement geometry defined in an inertial reference frame (I) with axes X_I, Y_I, Z_I , as illustrated in Figure 1. Let r_M^I and r_T^I denote the position vectors of the missile and the target, respectively. The relative position vector r_{MT}^I is defined as $r_{MT}^I = r_T^I - r_M^I$, and similarly the relative velocity vector is given by $\dot{r}_{MT}^I = V_T^I - V_M^I$.

The line-of-sight (LOS) rate vector, $\dot{\lambda}$, essential for proportional navigation guidance, is derived from the relative kinematics as:

$$\dot{\lambda} = \frac{r_{MT}^I \times \dot{r}_{MT}^I}{\|r_{MT}^I\|^2} \quad (1)$$

The guidance filter aims to accurately estimate r_{MT}^I and $\dot{r}_{MT}^I$ to compute this LOS rate.

2.2 Strapdown SAL Seeker Model

2.2.1 General Description and Operational Principle

A guided weapon equipped with a Semi-Active Laser (SAL) seeker operates by detecting laser energy reflected from a target. This process requires an external laser designator, which may be either co-located with the launch platform or positioned on a separate platform. The designator illuminates the target with a directed laser

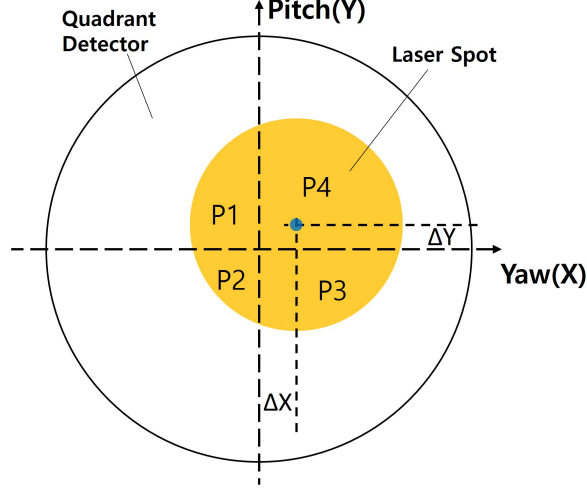


Fig. 2 Schematic of a four-quadrant photodiode detector and laser spot centroid displacement $(\Delta x, \Delta y)$ logic.

beam, and the emitted signal reflects off the target surface. The seeker receives this reflected signal and detects its energy distribution to extract the directional information necessary for steering the weapon toward the target.

Although various types of detectors can be applied to SAL seekers, those employing four-quadrant photodiodes are the most commonly used in practice due to their reliability and processing speed [8, 9]. As illustrated in Figure 2, a four-quadrant detector consists of four segmented photodiodes (P_1, P_2, P_3, P_4). Each segment independently detects the magnitude (or power) of the reflected laser signal. The angular information of the target is derived based on the relative intensity of the signal measured in each quadrant.

The normalized displacement of the laser spot centroid $(\Delta x, \Delta y)$ on the detector plane is calculated using the following equations:

$$\begin{aligned} \frac{\Delta y}{R_{\text{spot}}} &= \frac{(P_1 + P_4) - (P_2 + P_3)}{P_{\text{total}}} \\ \frac{\Delta x}{R_{\text{spot}}} &= \frac{(P_3 + P_4) - (P_1 + P_2)}{P_{\text{total}}} \end{aligned} \quad (2)$$

where $P_{\text{total}} = \sum_{i=1}^4 P_i$ represents the total received power sum. Here, R_{spot} denotes the effective radius of the laser spot. Since the radius of the laser spot is determined by optical factors—specifically the defocusing distance between the seeker lens and the detector plane—Eq. (2) provides an estimate of the spot's center position relative to the optical axis.

Finally, the pitch (σ_θ) and yaw (σ_ϕ) look angles toward the target in the body frame are approximated as:

$$\sigma_\phi \approx \frac{\Delta x}{f}, \quad \sigma_\theta \approx \frac{\Delta y}{f}$$

where f is the focal length of the seeker's optical system. These body-frame angles serve as the primary input for the coordinate transformation and subsequent filtering processes.

2.2.2 Coordinate Transformation and LOS Extraction

Since the seeker measures angles in the body frame, as shown in Figure 3, these must be transformed into the inertial frame to be compatible with the guidance filter. First, the unit direction vector pointing toward the target in the body frame, i_{MT}^B , is constructed from the look angles:

$$i_{MT}^B = \frac{1}{\sqrt{1 + \tan^2 \sigma_\phi + \tan^2 \sigma_\theta}} \begin{bmatrix} 1 \\ \tan \sigma_\phi \\ -\tan \sigma_\theta \end{bmatrix}$$

This vector is then transformed into the inertial frame using the coordinate transformation matrix C_B^I , which is obtained from the onboard Inertial Navigation System (INS):

$$i_{MT}^I = C_B^I i_{MT}^B.$$

Finally, the inertial LOS azimuth (λ_ϕ) and elevation (λ_θ) angles are extracted from the inertial unit vector components $i_{MT}^I = (i_x, i_y, i_z)$:

$$\begin{aligned} \lambda_\phi &= \tan^{-1}(i_y/i_x) \\ \lambda_\theta &= \tan^{-1}\left(-i_z/\sqrt{i_x^2 + i_y^2}\right) \end{aligned} \tag{3}$$

This formulation highlights the coupling between the seeker measurements and the INS attitude solution, characteristic of strapdown systems.

2.3 Guidance Filter Formulation

To estimate the LOS rate, we employ a Cartesian coordinate-based Kalman filter structure.

2.3.1 System Model

The state vector x_k is defined as the concatenation of relative position and velocity vectors:

$$x_k = (r_{MT}^I, \dot{r}_{MT}^I) \in \mathbb{R}^6$$

The discrete-time state transition equation is given by:

$$x_k = F_{k-1} x_{k-1} + w_{k-1} \tag{4}$$

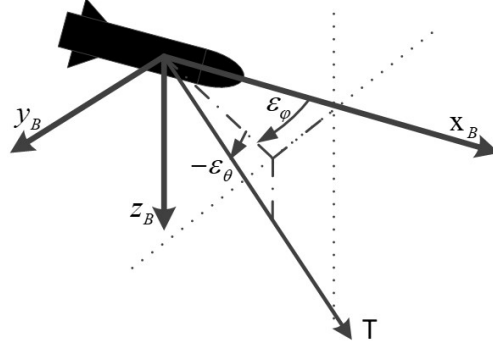


Fig. 3 Geometric relationship between missile body frame, target line-of-sight (LOS), and look-angle components for a strapdown configuration.

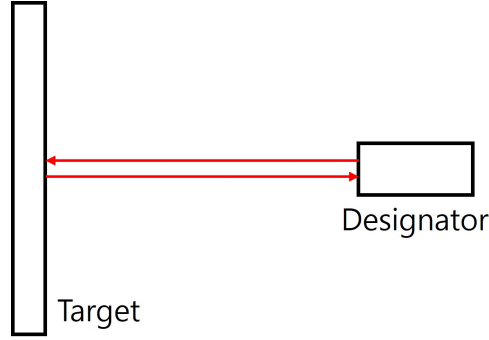


Fig. 4 Illustration of fundamental laser reflection mechanisms: Retroreflection.

where the state transition matrix F_{k-1} models the kinematics over the sampling interval Δt :

$$F_{k-1} = \begin{bmatrix} I_{3 \times 3} & \Delta t I_{3 \times 3} \\ 0_{3 \times 3} & I_{3 \times 3} \end{bmatrix}$$

The process noise w_{k-1} accounts for target acceleration uncertainties and is modeled as zero-mean Gaussian noise, $w_{k-1} \sim \mathcal{N}(0, Q_{k-1})$.

2.3.2 Measurement Model

The measurement vector z_k consists of the inertial LOS angles derived in Eq. (3) and the relative range r_m . The relative range is not directly measured by the SAL seeker. Instead, it is calculated using the missile position provided by the onboard navigation system and the estimated target position. The target position is assumed to be available from pre-launch target position and velocity information or from in-flight target updates provided through a data link. In the present study, the resulting relative-range information is assumed to be independent of the seeker angular measurements:

$$z_k = (\lambda_\phi, \lambda_\theta, r_m)$$

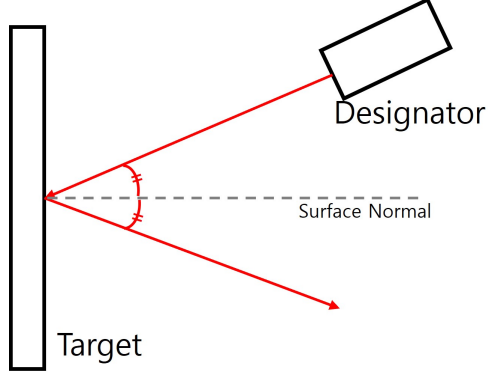


Fig. 5 Illustration of fundamental laser reflection mechanisms: Specular reflection.

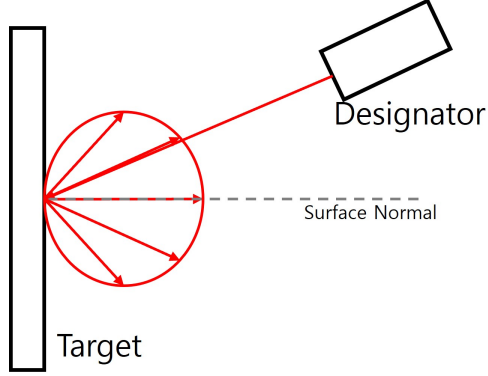


Fig. 6 Illustration of fundamental laser reflection mechanisms: Lambertian (diffuse) reflection.

The nonlinear measurement model $h(x_k)$ relates the state variables to the observations:

$$\begin{aligned}
 z_k &= h(x_k) + v_k \\
 &= \begin{bmatrix} \tan^{-1}(Y_{MT}^I/X_{MT}^I) \\ \tan^{-1}(-Z_{MT}^I/\sqrt{(X_{MT}^I)^2 + (Y_{MT}^I)^2}) \\ \sqrt{(X_{MT}^I)^2 + (Y_{MT}^I)^2 + (Z_{MT}^I)^2} \end{bmatrix} + v_k
 \end{aligned} \tag{5}$$

with $r_{MT}^I = (X_{MT}^I, Y_{MT}^I, Z_{MT}^I)$. In standard formulations, the measurement noise v_k is assumed to be zero-mean Gaussian, $v_k \sim \mathcal{N}(0, R_k)$. However, as analyzed in the subsequent section, this assumption is compromised in the presence of specular reflections.

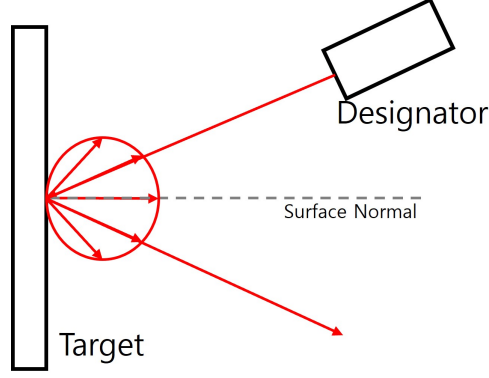


Fig. 7 Illustration of fundamental laser reflection mechanisms: Combined reflection on complex target surfaces.

3 Experimental Characterization of Measurement Disturbances

To investigate the validity of the Gaussian noise assumption and to characterize the actual measurement disturbances encountered by strapdown SAL seekers, we analyze the physical reflection mechanisms and present detailed results from outdoor vehicle experiments.

3.1 Reflection Mechanisms and Outlier Generation

Reflected laser signals can be broadly classified into three fundamental reflection mechanisms, with an additional combined-reflection case commonly observed on practical targets [8]:

1. *Retroreflection (Figure 4)*: The beam is directed back toward the source, which is characteristic of corner-cube reflectors but rare on standard targets.
2. *Specular Reflection (Figure 5)*: Also known as mirror-like reflection, the beam exits at an angle equal to the incident angle. This typically occurs on smooth surfaces such as glass windows or polished metallic bodies.
3. *Lambertian Reflection (Figure 6)*: The ideal diffuse reflection where energy is scattered uniformly in all directions, which is the standard assumption for stable tracking.
4. *Combined Reflection (Figure 7)*: Practical targets often contain multiple surface types within the laser spot. This mixed reflection condition can produce a non-uniform energy distribution on the detector.

In practical engagement scenarios, targets such as vehicles or ships often exhibit a mixture of these properties, referred to as *combined reflection* (Figure 7). As illustrated in Figure 7, when the laser spot illuminates a target containing both matte surfaces (Lambertian) and smooth components like grille or windows (Specular), a significant portion of the reflected energy tends to concentrate in the specular direction. This results in a non-uniform energy distribution on the four-quadrant detector, where



Fig. 8 Experimental setup test vehicle and Lambertian reference plates.

energy is heavily biased toward one or a few quadrants. Consequently, the estimated centroid shifts significantly, introducing large, outlier-type bias errors that violate the Gaussian noise assumption used in standard filters.

3.2 Outdoor Experiments: Setup and Methodology

Outdoor experiments were conducted to characterize these effects using a standard passenger van as a representative target. The objective was to observe how different surface materials and mixed reflectivity within the laser spot affect measurement accuracy.

3.2.1 Test Condition

To investigate the reflective characteristics of typical targets with various surface materials, a standard passenger van was used as the test target. Two types of tests were conducted:

- *Test 1 (Surface Material Dependency)*: Aimed to observe the reflective characteristics of individual target surfaces by comparing actual measurements with those obtained from a Lambertian reference plate.
- *Test 2 (Mixed Reflectivity)*: Designed to analyze how measurement results are affected when different reflectivities exist simultaneously within the same laser spot area.

Figure 8 shows the test setup including the vehicle and the Lambertian reflection plates. Among the reflection plates, the center one has a reflectivity of 50%, and the left one has a reflectivity of 18%, which were selected to emulate surfaces with different reflective intensities.



Fig. 9 Identification of target surface components (Window, Bonnet, Grille, Body, and Tire) for localized laser reflection measurements.

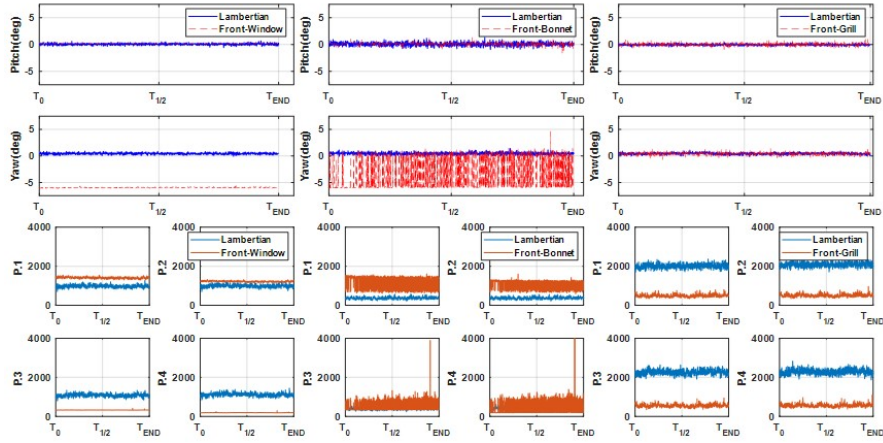


Fig. 10 Comparative analysis of look-angle measurements and quadrant signal intensities (P_1 – P_4) for various target orientations and surface materials (Front Window/Bonnet/Grille).

3.3 Analysis of Disturbance Characteristics

3.3.1 Test 1: Surface Material Dependency

Measurements were repeated at the same location while varying the target orientation (Front, Diagonal, Side) and specific surface components (Window, Bonnet, Body, Grille, Tire), as shown in Figure 9. The results were compared with the Lambertian baseline to identify bias sources.

Specular Dominance (Window, Bonnet & Diagonal-Body in Figure 10–12):

In the Front-Window, Diagonal-Window, Diagonal-Bonnet, and Diagonal-Body cases, the received signal intensity exhibited a distinct asymmetry due to high reflectivity and geometric orientation. Specifically, strong signals were observed in quadrants P_1 and P_2 , whereas signals in P_3 and P_4 were significantly weaker. This energy imbalance

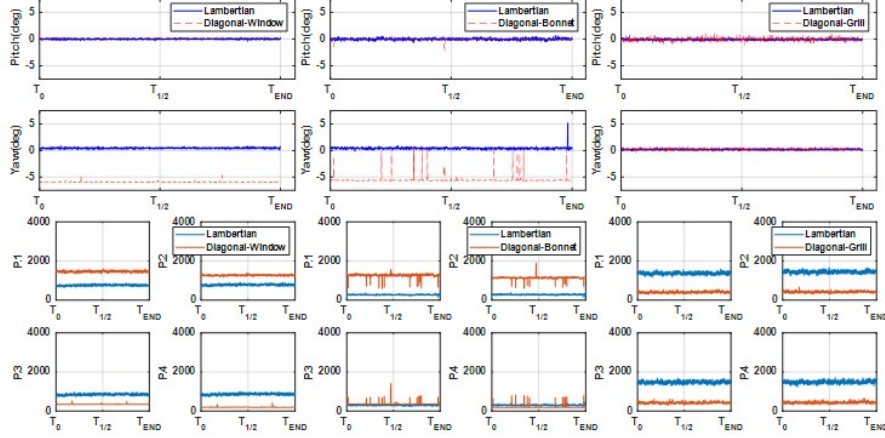


Fig. 11 Comparative analysis of look-angle measurements and quadrant signal intensities (P_1 – P_4) for various target orientations and surface materials (Diagonal Window/Bonnet/Grille).

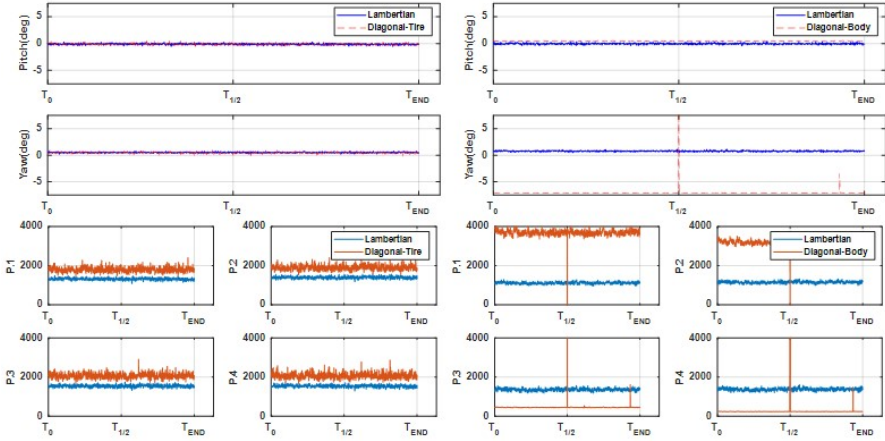


Fig. 12 Comparative analysis of look-angle measurements and quadrant signal intensities (P_1 – P_4) for various target orientations and surface materials (Diagonal Tire/Body).

resulted in large, deterministic bias-type errors in the estimated look angles. In the Front-Bonnet case, this bias was accompanied by large fluctuations in signal intensity, introducing additional measurement variance.

Grille Structures (Figure 10–11):

For the Front-Grille and Diagonal-Grille cases, the complex, lattice-like geometry caused laser scattering. Consequently, relatively weak signals were detected across all channels. Unlike the specular cases, this did not produce large directional biases but resulted in a significant increase in measurement noise variance (jitter) due to the low Signal-to-Noise Ratio (SNR).

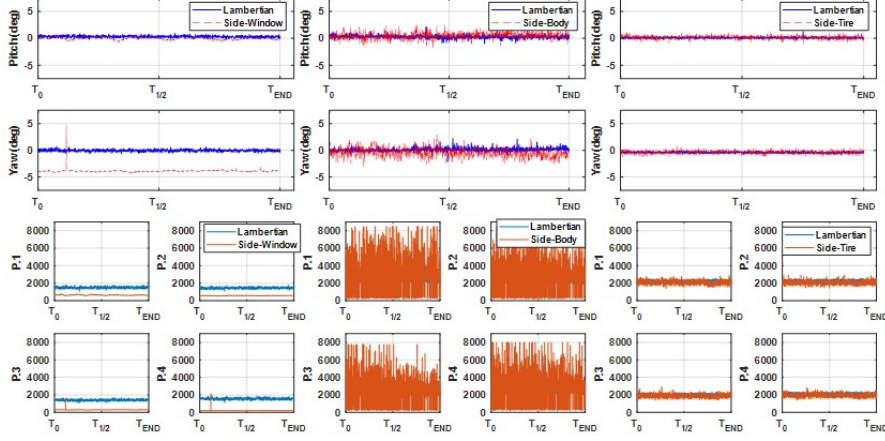


Fig. 13 Comparative analysis of look-angle measurements and quadrant signal intensities (P_1 – P_4) for various target orientations and surface materials (Side Window/Body/Tire).

Side Orientation (Figure 13):

The Side-Window case showed weak signals overall, but with greater attenuation in channels 3 and 4 compared to channels 1 and 2. This specific imbalance produced bias-type errors predominantly in the yaw direction. Conversely, Side-Body and Side-Tire cases exhibited noisy energy distributions with large intensity fluctuations across all channels, resulting in high angular variance without a dominant bias direction.

Lambertian-like Surfaces (Diagonal-Tire in Figure 12):

In contrast to other diagonal components, the Diagonal-Tire case showed a signal intensity increase that was evenly distributed across all channels. Consequently, the estimation error and variance levels were comparable to those of the Lambertian reference plate, confirming that non-reflective rubber surfaces behave nominally even at oblique angles.

The experiments confirm that strong localized reflections (specular) lead to bias errors, while structural scattering (grille) leads to high-variance noise. Under practical operating conditions, factors such as jitter from the laser designator, target motion, and variations in spot size cause the laser spot to traverse these diverse surfaces rapidly. As a result, the reflective characteristics within the spot vary dynamically. Consequently, these errors manifest not as constant offsets, but as transient, short-duration outliers that intermittently disturb the guidance loop.

In summary, for guided weapons employing SAL seekers, bias-type errors can arise principally from the reflective properties of the target. Therefore, it is essential either to design exception-handling logic that mitigates such effects or to develop filters that are robust to bias-type errors.

3.3.2 Test 2: Mixed Reflectivity Effects

To analyze the effect of non-uniform reflectivity within a single laser spot, a second series of tests was conducted where the laser spot illuminated boundaries between surfaces of different reflectivity. Table 1 summarizes the results of the four configurations:

Baseline (Case 1):

The laser spot was evenly distributed on a 50% reflectivity Lambertian plate. The measured pitch angle was -6.880° , which serves as the Lambertian reference baseline.

High Contrast (Case 2):

The spot covered half the 50% Lambertian plate and half a matte black surface (non-reflective, $< 10\%$ reflectivity). This configuration produced a measurement of -6.127° . This represents the largest deviation among all cases (approximately 0.75° bias), indicating that the centroid calculation is heavily skewed by the “missing” energy from the non-reflective half.

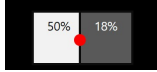
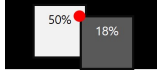
Low Contrast (Case 3):

The spot covered half the 50% plate and half an 18% Lambertian plate. Although the reflectivities differed, the measured value was -6.418° . Notably, the deviation in this case was smaller than in Case 2. This suggests that the *contrast* in reflection characteristics (Lambertian vs. Non-reflective) has a greater impact on estimation error than the simple difference in absolute reflectivity (High Lambertian vs. Low Lambertian).

Dominance Effect (Case 4):

When the spot was split 3:1 between the 50% and 18% plates, the error remained small (-6.576°), as the dominant surface area (75%) governed the centroid calculation

Table 1 Measurement deviations in pitch and yaw look angles under non-uniform reflectivity within the laser spot (Test 2).

No.	Target		Pitch	Yaw
1	Lambertian 50%		-6.880	-0.054
2	Half:Lambertian 50% Half:No Plate / black surface		-6.127	-0.781
3	Half:Lambertian 50% Half:Lambertian 18%		-6.418	0.078
4	Three-fourths:Lambertian 50% One-fourth:Lambertian 18%		-6.576	-0.352

These results confirm that mixed reflectance characteristics within a single spot—especially when there is a sharp contrast in surface properties—generate non-Gaussian errors. Along with the specular effects observed in Test 1, these phenomena justify the need for a robust filter capable of handling outlier-type disturbances.

3.4 Implication for Robust Filter Design

The experimental evidence collected from Tests 1 and 2 demonstrates that measurement noise in SAL seekers can exhibit strongly non-Gaussian characteristics in representative outdoor environments. The presence of specular reflections and mixed surface boundaries introduces transient biases and outlier-type error characteristics. The purpose of these experiments is to characterize the occurrence and qualitative behavior of such disturbances, rather than to identify a complete probabilistic disturbance model or to statistically estimate the bias magnitudes and durations used in the subsequent simulations. Instead, the experimental observations provide the physical motivation for evaluating the guidance filters under controlled finite-duration angular-bias conditions. Because conventional ℓ_2 -norm-based filters are highly sensitive to such outliers, a robust estimation framework is required. Accordingly, a Huber-based guidance filter is formulated in the following section.

4 Robust Guidance Filter Design

Based on the experimental analysis in Section 3, it is evident that strapdown SAL seeker measurements can be corrupted by non-Gaussian, outlier-type disturbances due to specular reflections. To address this challenge, we propose a Huber-based Guidance Filter (HGF). The primary objective of this filter is to robustly estimate the relative position r_{MT}^I and velocity $\dot{r}_{MT}^I$ required to compute the line-of-sight (LOS) rate $\dot{\lambda}$, specifically within the strapdown seeker dynamics framework defined in Section 2. Unlike conventional Extended Kalman Filters (EKF) that rely on the ℓ_2 -norm (Minimum Mean Square Error) and are inherently sensitive to outliers, the proposed HGF utilizes the Huber generalized cost function. This approach effectively down-weights anomalous measurements while maintaining EKF-like accuracy under nominal conditions, ensuring stable guidance command generation even when seeker measurements z_k are corrupted by specular reflections.

4.1 Linearization and Whitening Transformation

To apply the Huber estimation framework, we first reformulate the nonlinear estimation problem into a linear regression form. Consider the nonlinear measurement model defined in Eq. (5):

$$z_k = h(x_k) + v_k.$$

Let $\bar{x}_k$ and $\bar{P}_k$ denote the a priori state estimate and error covariance, respectively. Let $\delta_k = x_k - \bar{x}_k$ denote the a priori estimation error. By linearizing the measurement equation around $\bar{x}_k$, we obtain:

$$z_k \approx h(\bar{x}_k) + H_k(x_k - \bar{x}_k) + v_k,$$

where H_k is the Jacobian matrix of the measurement function with respect to the state vector $x_k = (r_{MT}^I, \dot{r}_{MT}^I)$. Specifically for the guidance problem, H_k has the following block structure:

$$H_k = \left. \frac{\partial h}{\partial x} \right|_{x=\bar{x}_k} = \begin{bmatrix} \partial \lambda_\phi / \partial r & 0_{1 \times 3} \\ \partial \lambda_\theta / \partial r & 0_{1 \times 3} \\ \partial r_m / \partial r & 0_{1 \times 3} \end{bmatrix}. \quad (6)$$

As shown in Eq. (6), the partial derivatives with respect to the velocity states are strictly zero. This indicates that the LOS rate information is inferred strictly from the temporal evolution of the position-dependent measurements. Consequently, any unsuppressed outlier in the position measurement (e.g., a glint-induced angle error) can propagate directly into the velocity estimate, destabilizing the guidance loop. To express the prior information and the new measurements in a common normalized regression form, we define the block-diagonal covariance matrix T_k and apply a whitening transformation:

$$T_k = \begin{bmatrix} R_k & 0 \\ 0 & \bar{P}_k \end{bmatrix}.$$

Using the Cholesky factorization $T_k = S_k S_k^T$, we define the whitened auxiliary variables as follows [25]:

$$d_k = S_k^{-1} \begin{bmatrix} z_k - h(\bar{x}_k) + H_k \bar{x}_k \\ \bar{x}_k \end{bmatrix}, \quad (7)$$

$$G_k = S_k^{-1} \begin{bmatrix} H_k \\ I \end{bmatrix}, \quad (8)$$

The first block of the transformed regression corresponds to the seeker measurements, whereas the second block represents the Gaussian prior-state information. This distinction is retained in the robust cost function introduced below. The transformed regression is constructed at the prior estimate ($\bar{x}_k$) and kept fixed during the IRLS iterations, while only the Huber weights are updated. Thus, IRLS is applied to a fixed first-order measurement approximation.

$$\xi_k = S_k^{-1} \begin{bmatrix} v_k \\ -\delta_k \end{bmatrix} = \begin{bmatrix} \xi_{z,k} \\ \xi_{p,k} \end{bmatrix}, \quad (9)$$

$$\xi_{z,k} = L_{R,k}^{-1} [z_k - h(\bar{x}_k) + H_k(x - \bar{x}_k)], \quad \xi_{p,k} = L_{P,k}^{-1} [\bar{x}_k - x] \quad (10)$$

$$R_k = L_{R,k} L_{R,k}^T, \quad \bar{P}_k = L_{P,k} L_{P,k}^T$$

Through this transformation, the estimation problem is converted into a standard linear regression form:

$$d_k = G_k x_k + \xi_k$$

where ξ_k is the normalized residual vector with identity covariance.

4.2 Huber-Based Cost Function and Optimization

The standard EKF measurement update is based on a quadratic cost under Gaussian prior and measurement assumptions. In the present problem, however, the non-Gaussian disturbances originate from the SAL seeker measurements rather than from the predicted state. Accordingly, the revised HGF retains the quadratic cost for the prior-state residual and applies the Huber loss only to the normalized seeker-measurement residuals:

$$J(x_k) = \frac{1}{2}(x_k - \bar{x}_k)^T \bar{P}_k^{-1}(x_k - \bar{x}_k) + \sum_{i=1}^m \rho(\xi_{z,k,i}(x_k)). \quad (11)$$

Here, the Huber function $\rho(\cdot)$ is defined as:

$$\rho(\xi_{z,k,i}) = \begin{cases} \frac{1}{2}\xi_{z,k,i}^2, & \text{if } |\xi_{z,k,i}| \leq \gamma \\ \gamma|\xi_{z,k,i}| - \frac{1}{2}\gamma^2, & \text{otherwise} \end{cases} \quad (12)$$

This hybrid formulation directly aligns with the physical reflection characteristics observed in the experiments of Section 3:

Gaussian Measurement Region (ℓ_2 -norm):

When $|\xi_{z,k,i}| \leq \gamma$, the corresponding seeker measurement retains a quadratic penalty. Therefore, under nominal Lambertian-reflection conditions, the measurement update approaches the conventional EKF update.

Outlier Measurement Region (ℓ_1 -norm):

When $|\xi_{z,k,i}| > \gamma$, the corresponding seeker-measurement component is assigned a linear penalty. This limits the influence of the corrupted angular channel while leaving the Gaussian prior term unchanged.

4.3 Iterative Solution via Re-weighted Least Squares

Minimizing the cost function $J(x_k)$ leads to an implicit equation. We solve this using the Iteratively Re-weighted Least Squares (IRLS) method. The optimal estimate at iteration $j + 1$ is given by:

$$\hat{x}_k^{(j+1)} = (G_k^T \Psi_k^{(j)} G_k)^{-1} G_k^T \Psi_k^{(j)} d_k, \quad \Psi_k^{(j)} = \text{blkdiag}(\psi_{z,k}^{(j)}, I_n) \quad (13)$$

where $\psi_{z,k}^{(j)} = \text{diag}(\psi_{k,1}^{(j)}, \dots, \psi_{k,m}^{(j)})$ is the weighting matrix. The weight of each seeker-measurement component is updated from the normalized measurement residual obtained at the previous iteration. No Huber weight is assigned to the prior-state residual. Consequently, a specular-reflection-induced residual spike reduces only the contribution of the affected seeker-measurement channel, while the predicted-state information remains fully retained.

$$\psi_{z,k,i}^{(j)} = \min(1, \gamma/|\xi_{z,k,i}^{(j)}|), \quad i = 1, \dots, m \quad (14)$$

After convergence, the final IRLS solution is adopted as the posterior state estimate $\hat{x}_k = \hat{x}_k^{(j)}$. The final measurement weights are then used to construct the equivalent measurement covariance as follows:

$$\begin{aligned} R_{k,eff} &= L_{R,k} (\psi_{z,k}^*)^{-1} L_{R,k}^T \\ K_k &= \bar{P}_k H_k^T (H_k \bar{P}_k H_k^T + R_{k,eff})^{-1} \end{aligned} \quad (15)$$

$$P_k = (I - K_k H_k) \bar{P}_k (I - K_k H_k)^T + K_k R_{k,eff} K_k^T$$

Here, $\psi_{z,k}^*$ denotes the final measurement-weight matrix after IRLS convergence.

Algorithm 1 Huber-based robust guidance filter (HGF) update procedure for outlier suppression.

Require: Measurement z_k , *a Priori* state $\bar{x}_k$, Covariance $\bar{P}_k$, Tuning parameter γ

Ensure: *a posteriori* state $\hat{x}_k$, covariance P_k , LOS Rate $\hat{\lambda}$

1: **1. Whitening Transformation:**

2: Compute Jacobian H_k using Eq. (6).

3: Construct combined covariance T_k and perform Cholesky decomposition $T_k = S_k S_k^T$.

4: Compute whitened matrices G_k and d_k using Eqs. (7)-(8).

5: **2. Iterative Reweighting (IRLS):**

6: Initialize $\hat{x}_k^{(0)} \leftarrow \bar{x}_k$, $j \leftarrow 0$.

7: **repeat**

8: Compute residuals: $\xi_k^{(j)} \leftarrow d_k - G_k \hat{x}_k^{(j)}$.

9: Update weights for each measurement i :

$$\psi_{k,i}^{(j)} = \min(1, \gamma/|\xi_{z,k,i}^{(j)}|)$$

10: Construct weighting matrix:

$$\Psi_k^{(j)} = \text{blkdiag}(\psi_{z,k}^{(j)}, I_n)$$

11: Update state estimate:

$$\hat{x}_k^{(j+1)} \leftarrow (G_k^T \Psi_k^{(j)} G_k)^{-1} G_k^T \Psi_k^{(j)} d_k$$

12: $j \leftarrow j + 1$

13: **until** Convergence ($\|\hat{x}_k^{(j)} - \hat{x}_k^{(j-1)}\| < \text{tol}$)

14: **3. Final Update:**

15: Update State: $\hat{x}_k \leftarrow \hat{x}_k^{(j)}$.

16: Compute $R_{k,eff}$ and K_k using Eq. 15

17: Update Covariance: $P_k \leftarrow (I - K_k H_k) \bar{P}_k (I - K_k H_k)^T + K_k R_{k,eff} K_k^T$.

18: Compute LOS Rate $\hat{\lambda}$ from $\hat{x}_k$ using Eq. (1).

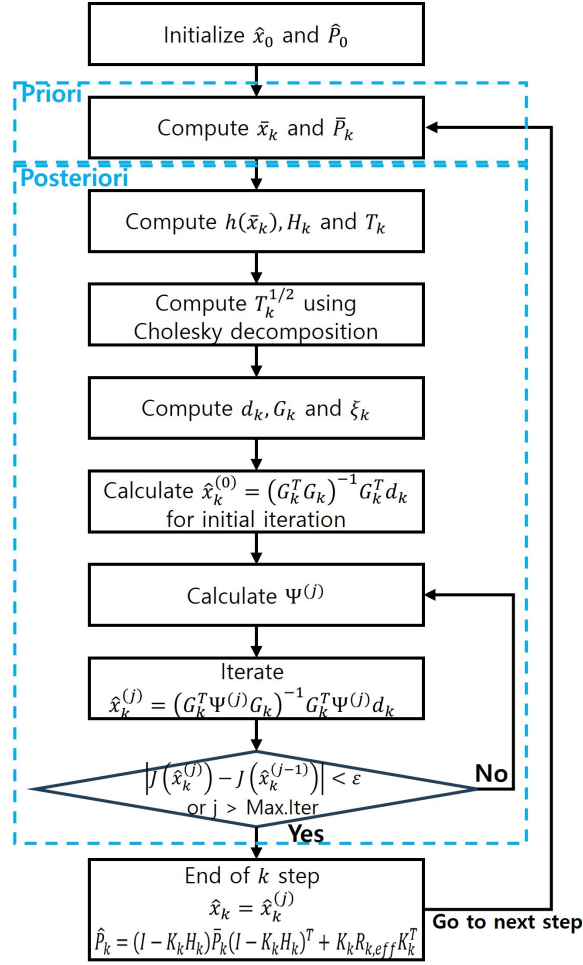


Fig. 14 Algorithmic flowchart of the proposed HGF illustrating the recursive state propagation and iterative measurement update stages.

4.4 Algorithm Implementation and Practical Interpretation

- *Scenario 1 (Selective Rejection)*: If a specular reflection occurs on a vertical window, only the elevation residual ($|\xi_{El}|$) spikes. The filter automatically calculates $\psi_{El} \ll 1$, effectively down-weighting the corrupted elevation component, while maintaining $\psi_{Az} \approx 1$ and $\psi_{Rng} \approx 1$ to continue updating the state using valid azimuth and range information.
- *Scenario 2 (Nominal Tracking)*: When tracking a diffuse target, all residuals remain within γ . Consequently, $\Psi_k \approx I$, and the HGF operates similarly to the standard EKF.

The implementation structure is visualized in Figure 14, and the detailed procedure is summarized in Algorithm 1. This algorithm is designed to maintain a smooth and bounded LOS-rate estimate even during intermittent specular disturbances.

As shown in the flowchart, the filter operates in the same *a priori* (propagation) structure as the conventional EKF, utilizing the standard state transition matrix F_{k-1} . However, the key distinction lies in the *a posteriori* (update) step. Unlike the conventional quadratic EKF measurement update, the HGF employs an iterative re-weighting loop.

The Huber cost function, which combines both the 1-norm and 2-norm terms, requires this iterative solution process until convergence.

1. *Initialization*: The iteration starts with the *a priori* estimate $\bar{x}_k$.
2. *Reweighting*: In each step, residuals ξ_k are checked against the threshold γ . If a specular reflection causes a residual spike (e.g., $|\xi| > \gamma$), the corresponding weight ψ is reduced.
3. *Update*: The state is updated using the adjusted weights.
4. *Convergence*: Once the iteration converges, the corresponding state is accepted as the final estimate $\hat{x}_k$.

Through this iterative process, the HGF provides a measurement-selective robust update for the considered strapdown SAL seeker problem. Its effectiveness under specular-reflection-induced disturbances is evaluated through the closed-loop simulations presented in the following section.

5 Performance Evaluation via Numerical Experiments

In this section, we evaluate the performance of the proposed Huber-based Guidance Filter (HGF) through high-fidelity numerical simulations. The objective is to demonstrate the HGF's robustness against the non-Gaussian measurement disturbances characterized in Section 3.

5.1 Simulation Configuration and Environment

To isolate the impact of seeker measurement disturbances, missile navigation errors and laser designator jitter are neglected, and only the detector noise of the laser seeker is considered in this study. Accordingly, the reported results demonstrate robustness to specular-reflection-induced measurement disturbances under the stated assumptions. The Huber weights are determined by the residual magnitude and do not distinguish the physical origin of the residual. The simulation focuses on a three-dimensional engagement involving a stationary ground target. The missile is guided using a Proportional Navigation Guidance (PNG) law to intercept the target. The finite-duration bias profiles are idealized test conditions motivated by the experimentally observed bias-like behavior, rather than parameters statistically inferred from the outdoor experiments. Case 1 examines the effect of disturbance duration, whereas Case 2 examines the effect of disturbance magnitude.

Table 2 summarizes the kinematic parameters and the filter configuration, including the process and measurement noise covariance matrices, Q and R . To ensure a fair

Table 2 Numerical simulation parameters and stochastic noise specifications for filter performance evaluation.

Parameter	Value
Initial Missile Position	(2500, 0, 3) m
Target Position	(0, 0, 0) m
Process Noise Covariance Q	diag(0.5, 0.5, 0.5, 0.5, 0.5, 0.5)
Measurement Noise Covariance R	diag(0.001 rad ² , 0.001 rad ² , 50 m ²)
Propagation / Update Rate	100 Hz/10 Hz
Huber Threshold (γ)	[0.5, 1, 1.5, 2]
Maximum Iterations ($j_{\max}$)	10
H_{∞} Filter (γ_{∞})	10

comparison, the standard Guidance Filter (GF based on EKF), the H_{∞} Guidance Filter (HinfGF), the improved maximum correntropy Kalman filter (IMCKF), and the proposed HGF utilize the same state transition model, nonlinear measurement model, process noise covariance, measurement noise covariance, initial conditions, update rates, and engagement conditions. The HinfGF is included as an additional robust benchmark and was evaluated with $\gamma_{\infty} = 10$, as a representative feasible setting. Under the present simulation configuration, no feasible Riccati-equation solution was obtained for $\gamma_{\infty} \leq 1$. The IMCKF in Ref. [27] is included as a representative non-Gaussian robust filtering benchmark, and its algorithm and parameter settings follow Ref. [27], with the kernel bandwidth set to $\sigma = 2$. In contrast, the proposed HGF employs an iterative re-weighted measurement update based on the Huber loss function, with the Huber threshold set to $\gamma = 1$. The sensitivity of the HGF performance to this threshold is separately evaluated in Section 5.3.3.

5.2 Performance Evaluation Scenarios

Two distinct cases are designed to stress-test the filters against the specular reflection characteristics observed in Tests 1 and 2.

- *Case 1 (Variable Duration)*: The magnitude of the pitch bias is fixed at 3° as a representative strong bias condition. The performance is analyzed as the duration of this bias increases from 0.2 s to 2.0 s in 0.2 s intervals.
- *Case 2 (Variable Magnitude)*: The duration of the bias is fixed at 1.0 s, while the magnitude of the yaw bias is gradually increased from 0.5° to 6.0° in 0.5° increments.

Table 3 summarizes the characteristics of the scenarios introduced for performance evaluation, including the magnitude and duration of the injected bias in each case. For each scenario condition, the identical bias onset, magnitude, and duration were applied to all filters to ensure a fair comparison. Performance metrics were gathered using 200 Monte Carlo Simulations (MCS) for each condition to provide statistical validity. The onset of the bias is triggered when the missile reaches half of its initial horizontal range ($R_0/2$).

Table 3 Injected disturbance parameters (onset, duration, and magnitude) for comparative performance stress-tests.

Case No.	Condition	Value
Case 1	Bias Onset	$R_0/2$
	Bias Duration (Δt)	$0.2 \sim 2.0$ s
	Bias Magnitude (b_{pitch})	3.0°
	Bias Magnitude (b_{yaw})	0.0°
Case 2	Bias Onset	$R_0/2$
	Bias Duration (Δt)	1.0 s
	Bias Magnitude (b_{pitch})	0.0°
	Bias Magnitude (b_{yaw})	$0.5 \sim 6.0^\circ$

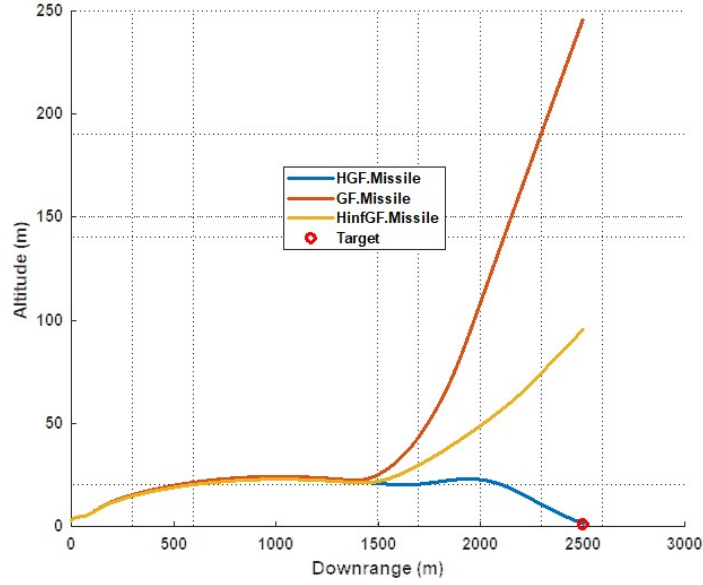


Fig. 15 Trajectory comparison of GF, HinfGF, and HGF under Case 1 disturbance.

5.3 Simulation Results and Discussion

Numerical simulations were conducted under the conditions defined in Table 3. This section analyzes the filter's performance in terms of trajectory stability, estimation accuracy, and statistical reliability.

5.3.1 Case-Specific Performance Analysis

Figures 15 and 16 show the representative simulation results for Case 1 with a 1.0 s bias duration, while Figures 17 and 18 present those for Case 2 with a 5.0° yaw

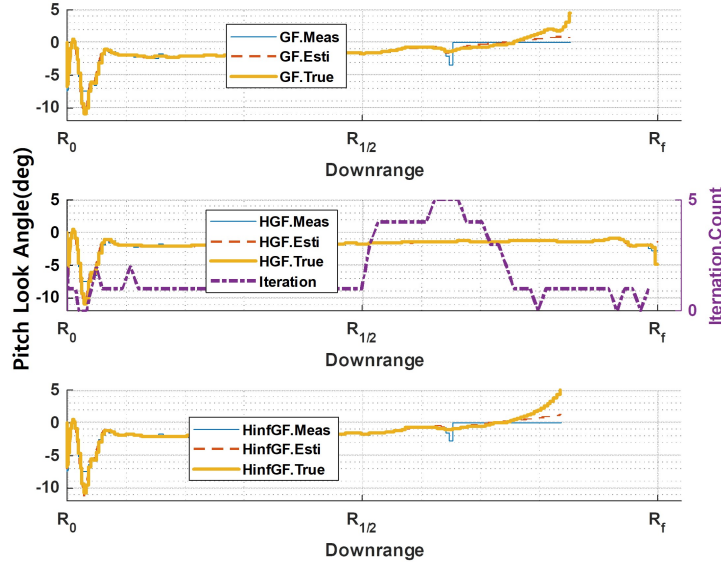


Fig. 16 Pitch look-angle comparison of GF, HinfGF, and HGF under Case 1 disturbance.

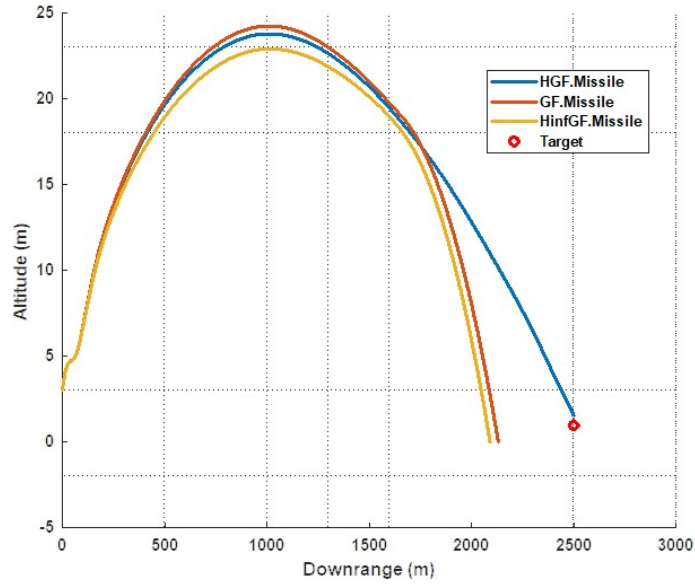


Fig. 17 Trajectory comparison of GF, HinfGF, and HGF under Case 2 disturbance.

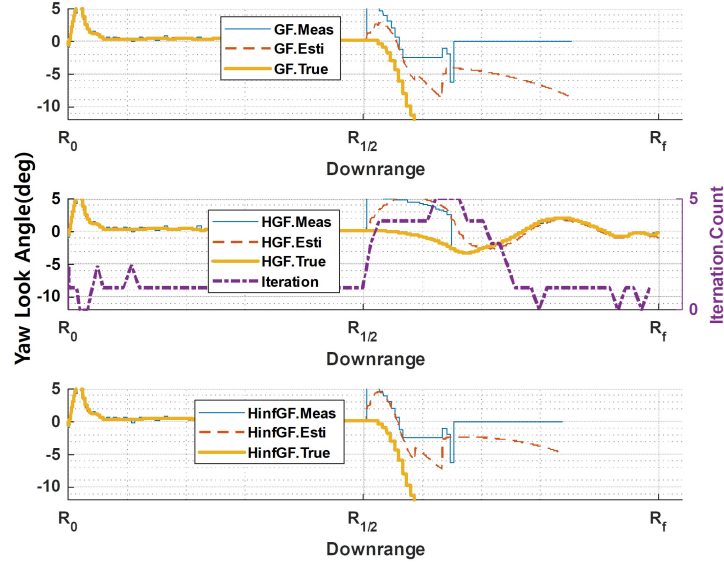


Fig. 18 Yaw look-angle comparison of GF, HinfGF, and HGF under Case 2 disturbance.

bias magnitude. The conventional GF, the H_∞ guidance filter (HinfGF), and the proposed HGF are compared in terms of trajectory stability and look-angle estimation behavior. For these representative cases, the attenuation level of the HinfGF was fixed at $\gamma_\infty = 10$. The HinfGF is used as a robust benchmark based on worst-case disturbance attenuation, whereas the HGF suppresses outliers through component-wise Huber re-weighting. These figures show the engagement trajectory, look-angle estimation accuracy, and HGF iteration behavior under specular-induced bias.

Trajectory and Look Angle Behavior:

In Case 1, where a pitch-direction bias is injected, the conventional GF exhibits a severe trajectory deviation after the onset of the biased measurement. The HinfGF reduces the sensitivity to the disturbance compared with the GF; however, its trajectory still deviates significantly from the desired intercept path when the biased measurement persists. In contrast, the proposed HGF maintains a stable trajectory toward the target by suppressing the influence of the biased look-angle measurement through the iterative Huber re-weighting process. In Case 2, where a yaw-direction bias is applied, the conventional GF again shows a large estimation deviation as the biased measurement is interpreted as a valid LOS change. The HinfGF suppresses part of this disturbance, but the remaining estimation error still causes a noticeable trajectory deviation. By contrast, the HGF maintains stable look-angle estimation and trajectory response, indicating that the proposed method is effective not only for pitch-direction disturbances but also for yaw-direction specular-induced bias. The look-angle responses further clarify the difference among the filters. In the GF, the

biased measurement is directly reflected in the estimated look angle, which causes an erroneous guidance command. The HinfGF provides a more conservative response than the GF, but it still exhibits noticeable tracking errors during the bias interval. The HGF estimate remains closest to the true look angle because the residuals associated with the biased measurement are down-weighted during the IRLS update. The increase in the HGF iteration count around the disturbance interval indicates that the filter detects the abnormal residual and activates the robust re-weighting mechanism.

Estimation Accuracy and Iterative Adaptation:

The iteration history of the HGF further explains how the proposed filter responds to abnormal measurements. The conventional GF is strongly affected by the biased measurements, leading to large estimation errors. The HinfGF with $\gamma_\infty = 10$ mitigates the disturbance to some extent, but residual deviations remain under persistent specular bias. In contrast, the proposed HGF estimates closely follow the true values by selectively down-weighting abnormal residuals. The increase in the iteration count during the disturbance interval confirms that the HGF activates its Huber-based re-weighting process to suppress the influence of specular glints.

LOS Rate Stability:

Since the guidance command is directly affected by the LOS rate estimate, the biased look-angle response of the conventional GF produces excessive LOS rate variations and destabilizes the guidance loop. The HinfGF with $\gamma_\infty = 10$ attenuates the disturbance to some extent, but residual LOS rate errors remain under the specular-induced bias. In contrast, the proposed HGF provides the smoothest and most consistent LOS rate estimate by selectively down-weighting abnormal residuals. This confirms its effectiveness in suppressing transient, non-Gaussian measurement bias caused by specular reflections.

In summary, the HinfGF with $\gamma_\infty = 10$ reduces the effect of biased measurements compared with the conventional GF; however, it still shows noticeable trajectory and look-angle deviations under persistent specular-induced disturbances. The proposed HGF achieves the most stable response because its Huber-based IRLS update selectively down-weights abnormal residual components rather than treating the disturbance only in a worst-case energy-bounded sense.

5.3.2 Statistical Robustness and Hit Probability

Because the estimated LOS rate is directly used in the guidance law, its accumulated estimation error is ultimately reflected in the terminal miss distance; therefore, miss distance and hit probability are used as the primary quantitative measures of closed-loop guidance robustness. To assess the statistical reliability of the proposed framework, 200 Monte Carlo simulations (MCS) were performed for each disturbance condition. Figure 19 summarizes the hit probability, defined as the percentage of runs with a miss distance of less than 1.0 m. The results compare the conventional GF, the H_∞ Guidance Filter (HinfGF), the IMCKF, and the proposed HGF under both bias-duration and bias-magnitude variations. The HinfGF result corresponds to the feasible attenuation setting of $\gamma_\infty = 10$ described in Section 5.1. The HinfGF improves

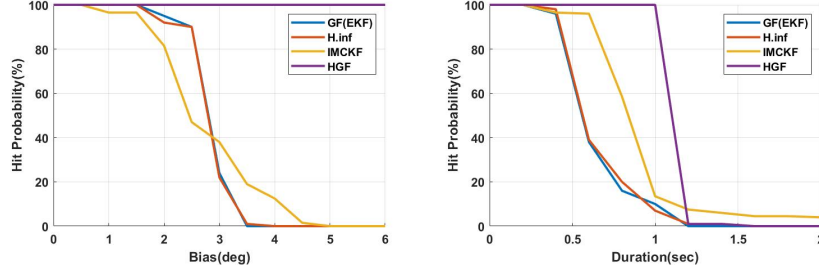


Fig. 19 Statistical intercept performance (Hit Probability, defined by a miss distance of less than 1.0 m) analysis as a function of bias duration and magnitude based on 200 Monte Carlo runs.

robustness compared with the GF, but its performance remains constrained by the feasible selection of γ_∞ .

Impact of Bias Duration (Case 1):

The statistical results reveal a significant performance threshold with respect to the disturbance duration. For Case 1, the pitch-direction bias magnitude is fixed at 3.0° , while the bias duration is increased from 0.2 s to 2.0 s. As shown in the right subplot of Figure 19, the conventional GF shows a sharp reduction in hit probability once the disturbance duration exceeds approximately 0.4 s, indicating that even a short persistent bias can destabilize the guidance loop. The HinfGF delays this degradation to some extent compared with the GF, but its hit probability still decreases when the biased measurement persists. The proposed HGF maintains near-perfect intercept performance for longer bias durations and provides a wider robust operating envelope. Although all filters eventually degrade under excessively prolonged bias, the HGF exhibits the highest resilience without requiring an aggressively tuned H_∞ attenuation bound.

Impact of Bias Magnitude (Case 2):

For Case 2, the bias duration is fixed at 1.0 s, while the yaw-direction bias magnitude is increased from 0.5° to 6.0° . As shown in the left subplot of Figure 19, the conventional GF maintains high hit probability only for relatively small bias magnitudes, but its intercept performance rapidly collapses as the bias becomes larger. The HinfGF improves robustness compared with the GF, but its performance remains strongly dependent on the selected attenuation level. Although a smaller γ_∞ can impose a more aggressive disturbance attenuation requirement, the feasible range of γ_∞ limits the achievable attenuation level. In contrast, the proposed HGF maintains consistently high hit probability over a wider bias-magnitude range by selectively down-weighting abnormal residual components through the Huber-based IRLS update.

Overall, Figure 19 confirms that the proposed HGF provides the most reliable intercept performance under both long-duration and large-magnitude specular disturbances. The conventional GF directly incorporates biased measurements into the state estimate, while the HinfGF attenuates disturbances in a worst-case energy-bounded

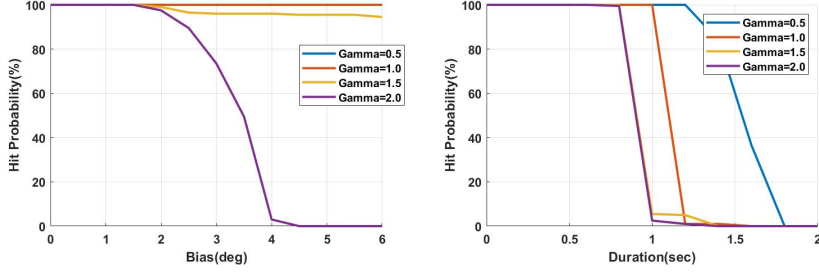


Fig. 20 Sensitivity of statistical intercept performance (Hit Probability, defined by a miss distance of less than 1.0 m) to the Huber threshold γ under bias-duration and bias-magnitude variations, based on 200 Monte Carlo runs.

sense and is subject to feasibility limitations depending on γ_∞ . By contrast, the HGF suppresses intermittent, non-Gaussian, and channel-dependent outliers through component-wise Huber re-weighting. This makes the proposed HGF more suitable for strapdown SAL seeker applications where specular reflections corrupt specific look-angle channels for short durations. The IMCKF also improves robustness against the non-Gaussian disturbance compared with the conventional GF. However, its hit probability decreases earlier than that of the HGF as the disturbance duration or magnitude increases.

5.3.3 Sensitivity to the Huber Threshold

A threshold of ($\gamma = 1$) was used as the baseline value in the preceding case studies. To investigate the influence of this parameter, additional simulations were conducted. All other filter parameters and engagement conditions were kept unchanged to isolate the effect of the Huber threshold.

The sensitivity analysis shows that smaller γ values provide stronger outlier suppression and higher robustness under severe reflection-induced disturbances. In the considered cases, $\gamma = 0.5$ and 1.0 maintained the highest hit probabilities, whereas performance degraded as γ increased to 1.5 and 2.0 . Accordingly, $\gamma = 1.0$ was retained as the nominal value for the main simulations.

5.3.4 Robustness Mechanism and Discussion

The superior performance of the proposed HGF is fundamentally attributed to its hybrid cost function. The conventional GF, based on the EKF framework, relies on the ℓ_2 -norm and therefore minimizes the squared innovation. Under nominal Gaussian measurement noise, this formulation is statistically efficient. However, when specular reflections generate large biased residuals, the quadratic penalty causes the outliers to dominate the cost function, pulling the state estimate toward the corrupted measurement.

The proposed HGF mitigates this limitation by employing the Huber cost function. For small residuals, the HGF retains the ℓ_2 -norm behavior and therefore preserves the nominal accuracy of the conventional EKF. For large residuals, however, the cost

function transitions to an ℓ_1 -norm-like penalty, which limits the influence of outliers. Through the IRLS update, this mechanism is implemented as a component-wise residual re-weighting process. As a result, abnormal residuals caused by specular reflections are selectively down-weighted, while the valid measurement components continue to contribute to the state update.

The comparison with the HinfGF further clarifies the different robustness mechanisms of the two approaches. The HinfGF improves robustness by attenuating the worst-case effect of energy-bounded disturbances on the estimation error. Therefore, it can reduce the sensitivity to biased measurements compared with the conventional GF. However, its performance depends strongly on the selected attenuation level γ_∞ , and the feasible range of this parameter limits how aggressively the disturbance attenuation can be imposed. In addition, the HinfGF does not explicitly identify which measurement component is corrupted, so persistent specular-induced bias can still be partially incorporated into the state estimate. The IMCKF provides robustness to non-Gaussian measurements through correntropy-based residual weighting. The results show that this mechanism also mitigates reflection-induced disturbances, but the HGF maintains a wider high-hit-probability region under the considered bias conditions.

In contrast, the HGF directly modifies the contribution of each whitened residual through the Huber weight. This component-wise suppression is particularly suitable for strapdown SAL seeker measurements, where specular reflections often corrupt only a specific look-angle channel for a short duration. Consequently, the proposed HGF achieves improved trajectory stability and LOS-rate estimation compared with the GF and HinfGF. In addition, it provides the highest statistical hit probability among the considered filters under the tested disturbance conditions.

6 Conclusion

This paper presented a robust Huber-based Guidance Filter (HGF) for strapdown Semi-Active Laser (SAL) seekers operating under specular reflection disturbances. Strapdown SAL seekers provide structural simplicity, low cost, and high-bandwidth guidance-loop characteristics, but their measurements can be severely degraded by non-Gaussian, outlier-type bias errors caused by specular reflections. Since conventional EKF-based Guidance Filters (GFs) rely on the ℓ_2 -norm and assume Gaussian measurement noise, they are highly sensitive to such biased measurements.

The first contribution of this study is the experimental characterization of SAL seeker measurement disturbances through outdoor vehicle experiments. The results showed that smooth and highly reflective surfaces, such as windows and polished body panels, can generate deterministic bias-type look-angle errors, whereas complex structures such as grilles can increase measurement variance due to weak and scattered reflected signals. In addition, mixed reflectivity within a single laser spot was shown to produce non-Gaussian measurement errors due to the non-uniform energy distribution on the four-quadrant detector. These observations confirm that practical SAL seeker measurements can contain intermittent and channel-dependent outliers that violate the Gaussian noise assumption.

The second contribution of this study applies a Huber M-estimation framework to the recursive LOS-rate estimation process used for strapdown SAL seeker guidance. Through whitening and component-wise reweighting, the resulting filter reduces the influence of abnormal residuals under the considered specular-reflection disturbance conditions while retaining nominal estimation performance. This structure allows valid measurement components to continue contributing to the state update even when a specific look-angle channel is corrupted.

Numerical simulations demonstrated that the proposed HGF provides superior robustness compared with the conventional GF, the HinfGF, and the IMCKF under the considered disturbance conditions. The conventional GF showed severe trajectory deviation, unstable look-angle response, and degraded hit probability when biased measurements were introduced. The HinfGF reduced the sensitivity to biased measurements compared with the GF; however, its performance depended on the feasible selection of the attenuation level γ_∞ , and residual trajectory deviations remained under persistent specular-induced bias. The IMCKF also improved robustness against non-Gaussian disturbances, but its hit probability degraded earlier than that of the HGF as the disturbance severity increased. In contrast, the proposed HGF maintained stable trajectory response, smooth LOS-rate estimation, and high hit probability under both long-duration and large-magnitude bias conditions. The prescribed bias profiles represent controlled parametric test conditions rather than a statistically identified stochastic disturbance model.

Overall, the proposed HGF provides a practical and structurally appropriate robust filtering solution for strapdown SAL seeker guidance in complex reflective environments. Rather than treating specular reflection only as a bounded worst-case disturbance, the HGF directly suppresses abnormal residual components at the measurement-update level. This makes the proposed approach effective for improving terminal guidance accuracy while preserving the advantages of low-cost strapdown seeker architectures. Future work will extend the proposed framework to more realistic engagement conditions, including maneuvering targets, laser designator jitter, navigation errors, and multi-sensor fusion. Hardware-in-the-loop simulations or flight-test-based validation will also be pursued to further verify the practical applicability of the proposed robust guidance filter.

Statements and Declarations

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Conflicts of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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